Practice session L6:
σ-point methods and Gaussian filters

Problem 1 – prediction using the cubature rule

The purpose with this problem is to study the distribution of σ-points in a situation where the variables have clear interpretations. Hopefully, this can help us to build up an intuition for why/how a σ-point method works.

Suppose that we have a state $\mathbf{x}_k = \begin{bmatrix} x_k \\ \dot{x}_k \end{bmatrix}$ that obeys a noise free constant velocity model with sampling time $T = 1$,

$$\mathbf{x}_k = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}_{k-1} \tag{1}$$

and that

$$\mathbf{x}_{k-1} \big| \mathbf{y}_{1:k-1} \sim \mathcal{N}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right). \tag{2}$$

a) What do you think $\mathbb{E}\left\{\mathbf{x}_k \big| \mathbf{y}_{1:k-1}\right\}$ is? What about $\mathrm{Cov}\left\{\mathbf{x}_k \big| \mathbf{y}_{1:k-1}\right\}$? Come up with qualified guesses, preferably using interpretations of x_k and $\dot{x}_k$, that you tell your group mates. Motivate your guesses.
Now compute the analytical values. Did you get the expected results?
Hint: $\mathrm{Cov}\{\mathbf{A}\mathbf{x}_{k-1} \big| \mathbf{y}_{1:k-1}\} = \mathbf{A}\mathbf{P}_{k-1|k-1}\mathbf{A}^T$.

b) Illustrate the mean and 1σ-ellipsoids of $p(\mathbf{x}_{k-1} \big| \mathbf{y}_{1:k-1})$ and $p(\mathbf{x}_k \big| \mathbf{y}_{1:k-1})$ in two different figures.
It is difficult to illustrate the ellipsoids in a precise manner, but just try to get the correlations and magnitudes (size of the ellipse) roughly correct (no need to compute the square root of $\mathbf{P}_{k|k}$).

Let us now study how we can perform prediction using the cubature rule.

c) Compute the σ-points $\mathcal{X}_{k-1}^{(1)}$, $\mathcal{X}_{k-1}^{(2)}$, $\mathcal{X}_{k-1}^{(3)}$ and $\mathcal{X}_{k-1}^{(4)}$, the associated weights $W_1, \ldots, W_4$ and illustrate $\mathcal{X}_{k-1}^{(1)}, \ldots \mathcal{X}_{k-1}^{(4)}$ in the figure with the corresponding ellipsoid. Recall that *the cubature rule* forms a set of $2n$ σ-points as follows:

$$\begin{aligned}
\mathcal{X}^{(i)} &= \hat{\mathbf{x}} + \sqrt{n}\mathbf{P}_i^{1/2}, & i &= 1, 2, \dots, n, \\
\mathcal{X}^{(i+n)} &= \hat{\mathbf{x}} - \sqrt{n}\mathbf{P}_i^{1/2}, & i &= 1, 2, \dots, n, \\
W_i &= \frac{1}{2n}
\end{aligned}$$

where $\mathbf{P}_i^{1/2}$ is the ith column of $\mathbf{P}^{1/2}$. For simplicity, I recommend that you use $\mathbf{P}_{k-1|k-1}^{1/2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. If you want, you can also ignore the scaling $\sqrt{2}$ in the description of the σ-points; this will make your calculation of the covariance incorrect but it will simplify your illustrations which I think is more important at the moment.

d) Set

$$\mathcal{X}_k^{(i)} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathcal{X}_{k-1}^{(i)} \tag{3}$$

for $i = 1, \dots, 4$ and illustrate them in the figure with the corresponding ellipsoids. Discuss what you observe and why.

e) Expressed using the notation from the lecture notes we have

$$\mathbf{f}(\mathbf{x}_{k-1}) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}_{k-1} \tag{4}$$

and the cubature rule suggests

$$\mathbb{E}\{\mathbf{x}_k \big| \mathbf{y}_{1:k-1}\} = \mathbb{E}\{\mathbf{f}(\mathbf{x}_{k-1}) \big| \mathbf{y}_{1:k-1}\} \approx \sum_{i=1}^{4} W_i \mathbf{f}(\mathcal{X}_{k-1}^{(i)}). \tag{5}$$

In this example, it holds that $\mathbf{x}_k = \mathbf{f}(\mathbf{x}_{k-1})$ since the model is noise-free, which means that $\mathcal{X}_k^{(i)} = \mathbf{f}(\mathcal{X}_{k-1}^{(i)})$ and

$$\mathbb{E}\{\mathbf{x}_k \big| \mathbf{y}_{1:k-1}\} \approx \sum_{i=1}^{4} W_i \mathcal{X}_k^{(i)}. \tag{6}$$

Use the cubature rule to approximate $\mathbb{E}\{\mathbf{x}_k \big| \mathbf{y}_{1:k-1}\}$. Compare with the analytic result from *b*).

f) If we use the notation $\boldsymbol{\mu}_x = \sum_{i=1}^{4} W_i \mathbf{f}(\mathcal{X}_{k-1}^{(i)})$, the cubature rule suggests the approximation

$$\begin{aligned}
\mathrm{Cov}\{\mathbf{f}(\mathbf{x}_{k-1}) \big| \mathbf{y}_{1:k-1}\} &\approx \sum_{i=1}^{4} W_i \left(\mathbf{f}(\mathcal{X}_{k-1}^{(i)}) - \boldsymbol{\mu}_x\right) \left(\mathbf{f}(\mathcal{X}_{k-1}^{(i)}) - \boldsymbol{\mu}_x\right)^T \\
&= \sum_{i=1}^{4} W_i \left(\mathcal{X}_k^{(i)} - \boldsymbol{\mu}_x\right) \left(\mathcal{X}_k^{(i)} - \boldsymbol{\mu}_x\right)^T.
\end{aligned} \tag{7}$$

Make use of the cubature rule to approximate $\mathrm{Cov}\{\mathbf{x}_k \big| \mathbf{y}_{1:k-1}\}$.

Problem 2 – approximation in the update step

An important aspect of the Gaussian filters discussed here is the type of Gaussian approximation used in the update step. Instead of directly computing the moments of

$$p(\mathbf{x}_k \big| \mathbf{y}_k, \mathbf{y}_{1:k-1}) \propto p(\mathbf{y}_k \big| \mathbf{x}_k)\mathcal{N}(\mathbf{x}_k; \hat{\mathbf{x}}_{k|k-1}, \mathbf{P}_{k|k-1}), \tag{8}$$

we first approximate the joint distribution

$$p(\mathbf{x}_k, \mathbf{y}_k \big| \mathbf{y}_{1:k-1}) \tag{9}$$

as Gaussian and then compute the conditional distribution $p(\mathbf{x}_k \big| \mathbf{y}_k, \mathbf{y}_{1:k-1})$ using the Lemma on conditional Gaussian densities.

A consequence of this is, for instance, that the posterior mean

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{P}_{xy}\mathbf{S}_k^{-1}(\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1}) \tag{10}$$

is a linear function of data. In spite of all our fancy σ-point methods to compute the moments of $p(\mathbf{x}_k, \mathbf{y}_k \big| \mathbf{y}_{1:k-1})$, we still express $\hat{\mathbf{x}}_{k|k}$ as a linear function of $\mathbf{y}_k$!

Let us look at these distributions in some examples and discuss what is going on here!

a) Suppose the measurement function $h(x)$ is highly nonlinear, as illustrated in Fig. 1. Let us also assume that we have very little measurement noise, that is,

$$y = h(x) + r, \tag{11}$$

where the variance of r is small. For this example, the joint distribution $p(x, y)$ cannot be accurately approximated as Gaussian, see Fig. 2 for an illustration of $p(x, y)$ and a Gaussian approximation with the same mean and covariance.

Try to sketch the posterior distribution $p(x \big| y)$ for the true distribution $p(x, y)$ and for its approximation $p_{\text{Gauss}}(x, y)$ as a function of x for different values of y, say $y = 0.15$, $y = 0.17$ and $y = 0.20$. Compare mean and covariance for the two versions.

Hint: As you may recall, it holds that $p(x \big| y) \propto p(x, y)$.

b) Discuss the differences if some of the parameters were different. For instance, what would happen if the prior (predicted) covariance was smaller? Or if the measurement noise was larger? Would you obtain a joint distribution $p(x, y)$ that you could approximate better or worse by a Gaussian density?

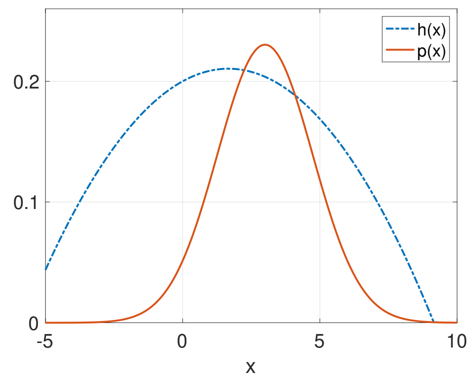

Figure 1: Here we illustrate the measurement function $h(x)$ together with the density $p(x)$.

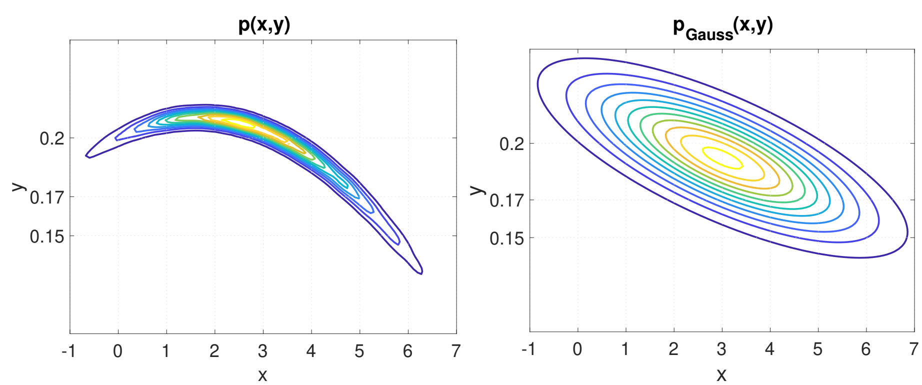

Figure 2: This figure contains the true density $p(x, y)$ [left] and a Gaussian approximation to that density [right].